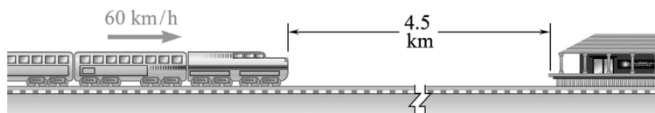


동역학 숙제 #2

(김병일 교수)

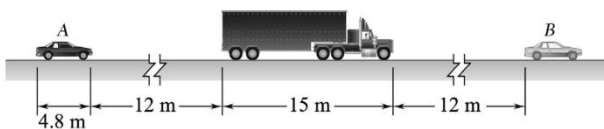
(Due in Two Weeks)

문제 1)



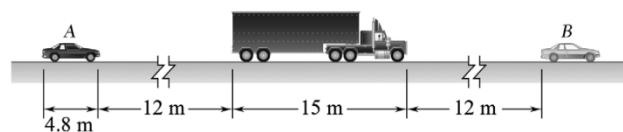
A commuter train traveling at 60 km/h is 4.5 km from a station. The train then decelerates so that its speed is 30 km/h when it is 0.75 km from the station. Knowing that the train arrives at the station 7.5 min after beginning to decelerate and assuming constant decelerations, determine (a) the time required for the train to travel the first 3.75 km, (b) the speed of the train as it arrives at the station, (c) the final constant deceleration of the train.

문제 2)



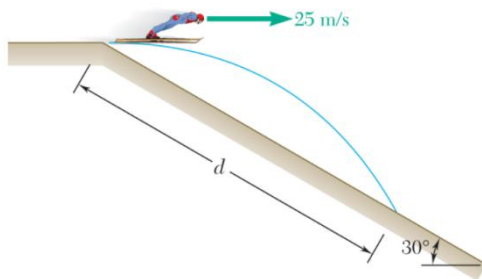
A car and a truck are both traveling at the constant speed of 50 km/h; the car is 12 m behind the truck. The driver of the car wants to pass the truck, i.e., he wishes to place his car at B, 12 m in front of the truck, and then resume the speed of 50 km/h. The maximum acceleration of the car is 1.5 m/s^2 and the maximum deceleration obtained by applying the brakes is 6 m/s^2 . What is the shortest time in which the driver of the car can complete the passing operation if he does not at any time exceed a speed of 75 km/h? Draw the $v - t$ curve.

문제 3)



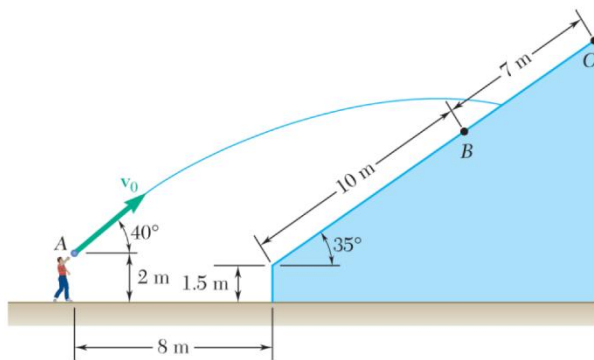
assuming that the driver of the car does not pay any attention to the speed limit while passing and concentrates on reaching position B and resuming a speed of 50 km/h in the shortest possible time. What is the maximum speed reached? Draw the $v - t$ curve.

문제 4)



A ski jumper starts with a horizontal take-off velocity of 25 m/s and lands on a straight landing hill inclined at 30° . Determine (a) the time between take-off and landing, (b) the length d of the jump, (c) the maximum vertical distance between the jumper and the landing hill.

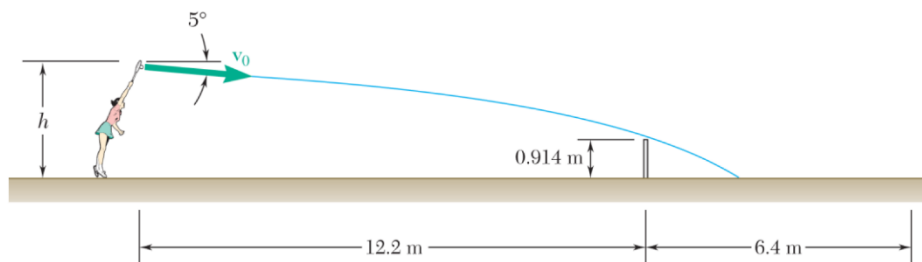
문제 5)



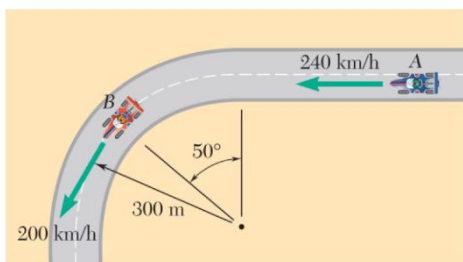
At halftime of a football game souvenir balls are thrown to the spectators with a velocity v_0 . Determine the range of values of v_0 if the balls are to land between Points B and C.

문제 6)

A tennis player serves the ball at a height $h = 2.5$ m with an initial velocity of v_0 at an angle of 5° with the horizontal. Determine the range for which of v_0 for which the ball will land in the service area which extends to 6.4 m beyond the net.

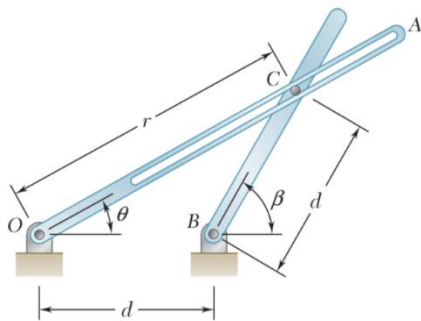


문제 7)



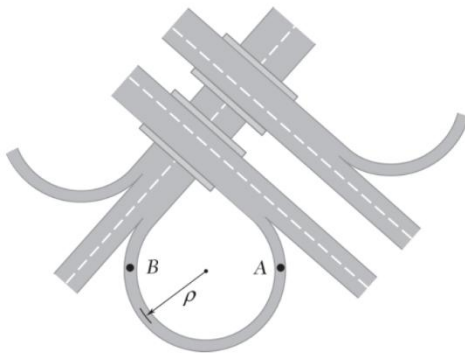
Racecar A is traveling on a straight portion of the track while racecar B is traveling on a circular portion of the track. At the instant shown, the speed of A is increasing at the rate of 10 m/s^2 , and the speed of B is decreasing at the rate of 6 m/s^2 . For the position shown, determine (a) the velocity of B relative to A, (b) the acceleration of B relative to A.

문제 8)



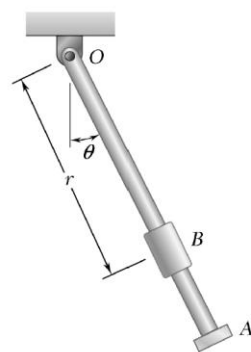
Pin C is attached to rod BC and slides freely in the slot of rod OA which rotates at the constant rate ω . At the instant when $\beta = 60^\circ$, determine (a) $\dot{r}$ and $\dot{\theta}$, (b) $\ddot{r}$ and $\ddot{\theta}$. Express your answers in terms of d and ω .

문제 9)



Determine the smallest radius that should be used for a highway if the normal component of the acceleration of a car traveling at 72 km/h is not to exceed 0.8 m/s^2 .

문제 10)



The oscillation of rod OA about O is defined by the relation $\theta = (2/\pi) (\sin \pi t)$, where θ and t are expressed in radians and seconds, respectively. Collar B slides along the rod so that its distance from O is $r = 625/(t + 4)$ where r and t are expressed in mm and seconds, respectively. When $t = 1 \text{ s}$, determine (a) the velocity of the collar, (b) the total acceleration of the collar, (c) the acceleration of the collar relative to the rod.